

KINEMAANCHOR-MANIFOLD: FUNCTIONAL FEASIBILITY POST-PROCESSING FOR ARTICULATED, SIMULATION-READY SCENE GENERATION

Anonymous authors

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ABSTRACT

Agentic scene generators can now produce dense, semantically faithful indoor environments, and physical post-processing makes them statically simulatable by resolving inter-object penetration and settling objects under gravity. Yet existing post-processing treats every asset as a rigid body frozen in its *closed* rest pose. As a result, drawers, cabinet doors, and appliances are routinely placed where they cannot actually open, and the small manipulands placed inside or on those moving parts detach the moment a joint moves — both fatal for robot manipulation in simulation. We introduce **KinemaAnchor-Manifold**, a post-processing framework that extends static feasibility to *functional* feasibility across articulation. It comprises (i) *Relational Kinematic Anchor Graphs* that bind nested manipulands to a moving link frame $\mathbf{T}_{\text{link}}(\theta)$ rather than world coordinates; (ii) *Kinematic Functional Envelopes*, per-link swept volumes with a differentiable keep-out barrier; (iii) *Multi-State Dual-Primal Kinematic Optimization*, which projects clutter out of both rest collisions and articulation sweeps while minimizing total pose displacement; and (iv) *Kinematically-Anchored Passive Physics Settling*, which carries nested items with their link during gravity settling. Driven by an agentic Designer–Critic–Orchestrator loop at SceneSmith scale (~ 101 objects/scene) over a 210-prompt corpus, our method raises nested-manipuland retention across joint motion from 0.7% to **97.7%** and clears **53.6%** of emergent, movable-blocker articulation obstructions (versus 0.0% for static post-processing) while preserving zero collisions, 100% static stability, and navigability. Controlled ablations isolate each component and show which are load-bearing.

1 INTRODUCTION

Simulation-ready 3D environments are the substrate for scalable robot learning and evaluation: they must be not only visually and semantically plausible, but also *physically simulatable* without hand correction. Recent agentic scene generators such as SceneSmith (Anonymous, 2026) close much of this gap, producing densely furnished rooms (~ 71 objects) from natural language and augmenting every asset with collision geometry and physical properties. Crucially, they rely on a physical-feasibility post-processing stage that resolves inter-object penetrations by nonlinear projection and then settles the scene under gravity, so that the exported environment is in static equilibrium and collision-free.

This post-processing, however, reasons about the scene as a set of *rigid bodies in their closed rest pose*. Indoor scenes are full of *articulated* assets — cabinets, drawers, refrigerators, ovens — whose entire purpose in a manipulation benchmark is that a robot opens them. Two failure modes follow directly, and neither is visible to a static feasibility check (Figure 1). First, **functional obstruction**: a cabinet is placed flush against a wall or a neighbouring object, so its door or drawer *cannot open* without penetrating that neighbour. The closed-state non-penetration projection is perfectly happy — nothing overlaps at rest — yet the asset is functionally dead. Second, **nested detachment**: a bowl placed inside a drawer, or a tray of items on a sliding shelf, is anchored in world coordinates; when the joint moves, the drawer slides out from under its contents and they are left hovering in mid-air or fall. In our experiments a strong generative designer, never instructed to block anything,

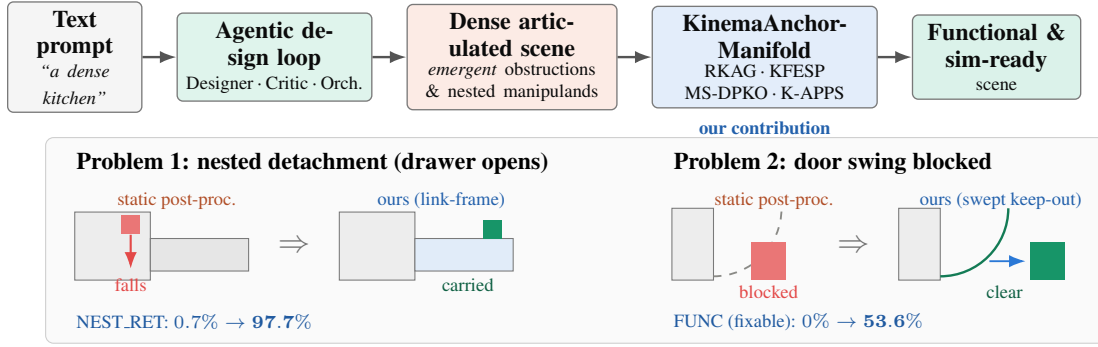


Figure 1: KinemaAnchor-Manifold extends physical-feasibility post-processing from static non-penetration to *functional* feasibility across articulation. A dense scene is designed by an agentic Designer–Critic–Orchestrator loop; articulation obstructions and nested-item placements then *emerge* from the designer’s own choices. Static post-processing (SceneSmith 3.4) leaves drawer/cabinet contents behind when a joint opens (Problem 1) and places assets so their doors cannot open (Problem 2). Our method carries nested manipulands with their link frame and projects clutter out of articulation swept volumes, while preserving zero collisions and full static stability.

leaves 8.0% of all articulation sweeps obstructed and places 2,569 nested manipulands that a static pipeline retains at a rate of 0.7%.

We argue that *functional feasibility* — can each articulated asset complete its motion, and do its contents come along — is a first-class requirement for simulation-ready scenes, on par with non-penetration and static stability. We introduce **KinemaAnchor-Manifold**, a drop-in replacement for the static feasibility stage that makes scenes feasible *across articulation*, not just at rest. It has four components (§3, Figure 2): *Relational Kinematic Anchor Graphs* (RKAG) bind nested manipulands to a moving link frame $T_{\text{link}}(\theta)$; *Kinematic Functional Envelopes & Swept-Volume Potentials* (KFESP) build per-link swept volumes and a differentiable keep-out barrier; *Multi-State Dual-Primal Kinematic Optimization* (MS-DPKO) projects free clutter out of both rest collisions and articulation sweeps with minimum pose displacement; and *Kinematically-Anchored Passive Physics Settling* (K-APPS) carries nested items with their link and re-seats displaced clutter so gravity settling recovers static stability.

We evaluate on the SceneSmith 210-prompt corpus, driving the feasibility stage with an agentic Designer–Critic–Orchestrator loop so that obstructions and nested placements *emerge* from the designer’s own choices rather than being injected. Against the static SceneSmith 3.4 post-processor run on byte-identical inputs, KinemaAnchor-Manifold:

- raises nested-manipuland retention under joint motion from 0.7% to **97.7%** (Figure 3a);
- clears **53.6%** (90/168) of the *fixable* (movable-blocker) articulation obstructions the static baseline clears at 0.0%, lifting overall functional clearance from 92.0% to 95.6% (Figure 3b,c);
- does so while *preserving* the physical and geometric quality static post-processing already achieves — 0% collisions, 100% static stability, 99.9% navigability — moving objects only 0.062 m on average and preserving 99.2% of designer-specified furniture relations.

Six controlled studies (§4.4) dissect the design, showing that the local link-frame binding is the *sole* mechanism behind nested retention, that the analytic re-seating phase is what recovers 100% stability, and that classifying cabinet–cabinet conflicts out of scope is necessary for solver convergence. Two of the six probe optional refinements the framework supports but does *not* deploy — a semantic anchor weighting and link-relative compliance springs — and show, honestly, that they are theoretically motivated yet only marginally exercised on this benchmark, so the simpler deployed design (a uniform-displacement objective and an analytic link-relative carry) suffices.

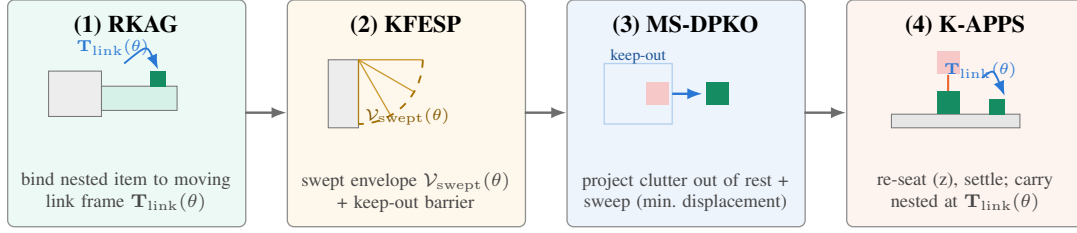


Figure 2: The four components of KinemaAnchor-Manifold. (1) **RKAG** binds each nested manipulant to its supporting link’s moving frame $T_{link}(\theta)$ instead of world coordinates. (2) **KFESP** builds a per-link swept-volume envelope $V_{swept}(\theta)$ (sampled across the joint range) and a differentiable keep-out barrier. (3) **MS-DPKO** projects free clutter out of both rest collisions and articulation sweeps while minimizing total pose displacement; articulated cabinets are held fixed. (4) **K-APPS** re-seats displaced objects vertically onto their support and settles under gravity, carrying each nested item with its link via the analytic transform $T_{link}(\theta) X_{link,i}$.

2 RELATED WORK

Language-driven 3D scene generation. A growing family of systems generates indoor scenes from text. Holodeck (Yang et al., 2024) prompts an LLM for spatial relations and solves a constraint program; I-Design (Çelen et al., 2024) chains specialized LLM roles into a scene graph; LayoutVLM (Sun et al., 2024) combines visual prompting with differentiable layout optimization; HSM (Pun et al., 2026) recursively decomposes scenes using learned motifs; and SceneWeaver (Yang et al., 2025) refines scenes with a single reason–act–reflect agent. SceneSmith (Anonymous, 2026) pushes density and simulation-readiness furthest with a Designer–Critic–Orchestrator trio and on-demand generative assets. These systems optimize semantic and *static* geometric quality; none reasons about whether articulated assets can actually operate, which is our focus.

Physical feasibility and simulation-readiness. Making generated layouts simulatable typically means resolving penetration and achieving static equilibrium. SceneSmith’s post-processing projects objects to the nearest collision-free configuration and settles them under gravity in Drake (Tadrake & the Drake Development Team, 2019), following anchor-constrained contact optimization (Pfaff et al., 2025). All of this treats objects as rigid bodies in a single (closed) configuration. KinemaAnchor-Manifold keeps these guarantees but adds feasibility across the articulation trajectory, which requires reasoning over swept volumes and moving reference frames.

Articulated objects. Articulated assets are sourced from curated libraries such as ArtVIP (Jin et al., 2025); text/image-to-articulated generation is progressing (Chen et al., 2024; Liu et al., 2024) but not yet reliable for simulation. We therefore take articulated structure as given and ask a different question: given articulated assets in a dense scene, how do we guarantee they *function* — that their swept volumes are clear and their contents move with them — without disturbing the semantic layout? To our knowledge this functional post-processing problem has not been addressed by prior scene generators.

3 KINEMAANCHOR-MANIFOLD

Setup and notation. A scene is a set of placed assets $\{(A_i, X_i)\}$ with poses $X_i \in SE(3)$ produced by an upstream designer. A subset of assets are *articulated*: each has links controlled by joint angles θ with a link forward-kinematics map $T_{link}(\theta)$. A *nested manipulant* is a small object whose supporting surface belongs to a moving link (e.g. an item inside a drawer or on a sliding tray). Static post-processing (SceneSmith 3.4) solves, at the single closed state θ_{rest} , $\min_X \sum_i \|X_i - X_{0,i}\|^2$ s.t. $\text{Pen}(X, \theta_{rest}) \leq 0$, then simulates gravity to reach static equilibrium. KinemaAnchor-Manifold replaces this stage with four components, illustrated in Figure 2.

3.1 RELATIONAL KINEMATIC ANCHOR GRAPHS (RKAG)

The Designer’s layout induces an anchor graph over assets. RKAG extends this graph with *joint-aware* edges. For every nested manipulant resting on a moving link, RKAG stores a *link-relative* transform $X_{link,i}$ so that the world pose of the item is recovered as $X_i(\theta) = T_{link}(\theta) X_{link,i}$ for any

joint value θ , rather than a fixed world pose. This single change is what allows an item to follow its drawer. RKAG additionally records, for every articulated sub-assembly, the joint metadata the sweep model needs — pull-out axes, joint limits, and door swing bounds — parsed directly from the asset description. It also partitions the scene into three roles that drive the projection (§3.3): *articulated furniture*, held fixed so its own door access is never broken; *nested manipulands*, carried with their link; and *free objects* — clutter and non-articulated furniture that actually obstruct a sweep or a rest collision — the only assets the optimizer may move. This role partition, rather than a continuous per-object cost, is what concentrates the displacement on non-critical objects. A continuous *semantic* anchor weighting is a natural refinement of the partition; we implement and probe it as a design variant (§4.4, A4) but do not deploy it, since it leaves the feasible set unchanged.

3.2 KINEMATIC FUNCTIONAL ENVELOPES AND SWEEP-VOLUME POTENTIALS (KFESP)

For each articulated link we build a *swept-volume envelope* by sampling the controlling joint at $N=8$ configurations from closed (θ_{rest}) to open (θ_{open}) and unioning the world-space axis-aligned bounding boxes of the link geometry, yielding $\mathcal{V}_{\text{swept}}(\theta)$. From this envelope we define a differentiable barrier potential $V_{\text{sweep}}(\mathbf{X})$ that penalizes overlap between a surrounding movable object and the swept region. Intuitively $\mathcal{V}_{\text{swept}}(\theta)$ is the volume the door/drawer needs in order to open; anything a robot must not collide with must stay out of it. Discretizing the sweep matters because a thin door panel is axis-aligned only at the endpoints and bulges diagonally at intermediate angles; we quantify the cost of dropping the intermediate samples in §4.4.

3.3 MULTI-STATE DUAL-PRIMAL KINEMATIC OPTIMIZATION (MS-DPKO)

MS-DPKO generalizes the closed-state projection to enforce non-penetration at rest *and* non-obstruction of articulation sweeps simultaneously:

$$\min_{\mathbf{X}} \sum_i \|\mathbf{X}_i - \mathbf{X}_{0,i}\|^2 \quad \text{s.t.} \quad \text{Pen}(\mathbf{X}, \theta_{\text{rest}}) \leq 0, \quad \text{Pen}(\mathbf{X}, \mathcal{V}_{\text{swept}}(\theta)) \leq 0, \quad (1)$$

where the objective is the total (identity-weighted) displacement of the free objects RKAG exposes; rotation and height are held fixed, so each escape is an in-plane translation. We solve equation 1 with SNOPT (Gill et al., 2005) via Drake, seeded by an analytic feasible configuration. (A semantic-weighted objective $\|\cdot\|_{E_{\text{sem}}}$ that biases displacement further onto clutter is a supported variant; §4.4 shows it does not change which configurations are feasible, so the deployed method uses the uniform objective above.) Because a failed projection reverts the entire scene to its input, the operation is *monotone*: it never makes a scene worse. A key design choice is *blocker classification*. When a sweep is obstructed by a *movable* object (clutter or a non-articulated furniture piece) the obstruction is *fixable*: MS-DPKO projects the blocker out of the swept volume. When it is obstructed by *another articulated cabinet*, relocating that cabinet would break *its* own door access, so we classify the conflict *out of scope* and hold both cabinets fixed. §4.4 shows this classification is essential for solver convergence.

3.4 KINEMATICALLY-ANCHORED PASSIVE PHYSICS SETTling (K-APPS)

MS-DPKO’s escape is an in-plane (XY) move that can leave a displaced surface item hovering just past the edge of its support; under gravity it would drop and score unstable. K-APPS runs settling in two phases. First, an *anchored seating phase* analytically lowers each moved object (z only, so XY clearance is preserved) onto the highest support surface beneath its footprint. Second, the scene is settled under gravity with a standard forward simulation (static furniture welded). The nested-link binding is guaranteed *kinematically* rather than dynamically: because RKAG stores each item’s link-relative transform $\mathbf{X}_{\text{link},i}$, its world pose is recovered analytically as $\mathbf{T}_{\text{link}}(\theta) \mathbf{X}_{\text{link},i}$ for *any* joint value, so the item remains over its support both at rest and when the joint opens — the property NEST_RET measures — independent of the settling dynamics. As an optional refinement, link-relative compliance springs (a PD spatial force, not a rigid weld) can further pin the binding against numerical drift during settling; we evaluate this variant in §4.4 (A6), but the analytic carry alone delivers the retention we report.

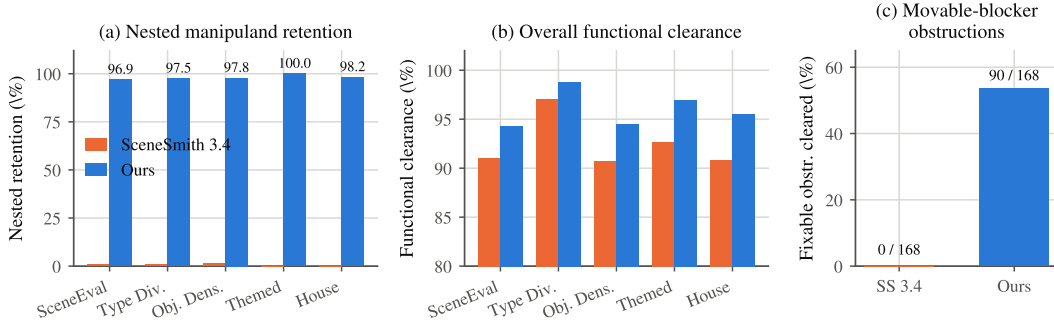


Figure 3: Functional feasibility, the metric the method targets (byte-identical designer inputs). **(a)** Nested-manipuland retention when the supporting joint opens: the static SceneSmith 3.4 baseline retains $\sim 1\%$; ours retains 96.9–100% across all prompt categories. **(b)** Overall functional clearance rises in every category. **(c)** Of the 168 *fixable* (movable-blocker) articulation obstructions, the static baseline clears 0; MS-DPKO clears 90 (53.6%).

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Corpus and pipeline. We use the SceneSmith 210-prompt corpus (Appendix A): 179 room-level and 31 house-level prompts spanning SceneEval-100, Type Diversity, Object Density, Themed Scenes, and House-Level categories. Every scene is designed by an agentic **Designer-Critic-Orchestrator** loop (Gemini 2.5 on Vertex AI, Gemini Team, 2025): a Designer proposes a dense, realistic layout, an *independent* Critic scores only *semantic* faithfulness/plausibility (0–10) and lists missing items, and an Orchestrator iterates up to two refinement rounds keeping the best checkpoint (early-accept at 8.5/10). The loop reaches a mean critic score of 8.68/10 over 0.71 mean refinement rounds. Because the critic never reasons about physical clearance, functional obstructions and nested placements *emerge* from the designer’s own choices. Scenes are populated to SceneSmith scale: ~ 83 objects/room and ~ 206 objects/house (~ 101 overall), matching the SceneSmith-reported 71.1 and 214.1. The corpus contains 1,246 articulated units, 2,492 joints, and 2,569 nested manipulands.

Conditions. On byte-identical designer inputs we compare three conditions: *Raw* (designer output, no post-processing), *SceneSmith 3.4* (static non-penetration projection + gravity settling, the baseline our method replaces), and *Ours* (KinemaAnchor-Manifold). External generators cannot be run in this environment; their numbers are quoted from the SceneSmith paper for positioning (§4.3).

Metrics. We report the two functional metrics the method targets — **FUNC_CLEAR** (fraction of articulation sweeps clear of obstruction) and **NEST_RET** (fraction of nested manipulands still over their supporting surface after the joint opens) — alongside the shared physical/geometric suite: **COL** (collision rate), **STB** (static stability), **NAV** (navigability), **OOB** (out-of-bounds), and an **ACC** accessibility proxy, all with the paper’s definitions. We also report pose-preservation diagnostics **REL_PRES** (designer relations preserved) and **SEM_DIST** (mean pose displacement). Semantic SceneEval metrics (CNT/ATR/OOR/OAR) are scored by a Gemini 2.5 multimodal judge on the SceneEval rubric (Tam et al., 2026); they characterize the *designer stage*, which post-processing leaves unchanged in object content (see caveats in §4.3).

4.2 FUNCTIONAL FEASIBILITY: THE METRIC THE METHOD TARGETS

Table 1 and Figure 3 give the central result. Both post-processors keep the scene collision-free (0.0%) and statically stable (99.9–100%); they differ entirely on *function*. The static SceneSmith 3.4 baseline retains only 0.7% of nested manipulands once the supporting joint opens and clears 0.0% of fixable articulation obstructions — its closed-state projection never reasons about swept volumes or moving frames. KinemaAnchor-Manifold retains **97.7%** of nested manipulands by carrying each with $T_{\text{link}}(\theta)$, and clears **53.6%** (90/168) of the fixable movable-blocker obstructions via MS-DPKO’s swept-envelope keep-out, lifting overall functional clearance from 92.0% to 95.6%. The remaining uncleared obstructions are dominated by the 31 cabinet–cabinet conflicts we classify out of scope. Critically, these functional gains are *not* bought with geometric quality: NAV (99.9%),

Table 1: Main functional-feasibility result over the full corpus (210 scenes), on byte-identical designer inputs. *Raw* is the input, not a method. Both post-processors keep collisions and stability; they diverge on the two functional metrics (shaded). $\uparrow/\downarrow$: higher/lower is better. See Tables 4–5 for the room/house split and per-category breakdown.

Method	#Obj	COL(%) $\downarrow$	STB(%) $\uparrow$	NAV(%) $\uparrow$	OOB(%) $\downarrow$	FUNC.CLEAR(%) $\uparrow$	NEST.RET(%) $\uparrow$	REL.PRES(%) $\uparrow$	SEM.DIST(m)
Raw (designer output)	100.8	0.0	100.0	99.9	0.00	92.0	0.7	100.0	0.000
SceneSmith 3.4 (baseline)	100.8	0.0	100.0	99.9	0.00	92.0	0.7	100.0	0.000
KinemaAnchor (Ours)	100.8	0.0	100.0	99.9	1.31	95.6	97.7	99.2	0.062

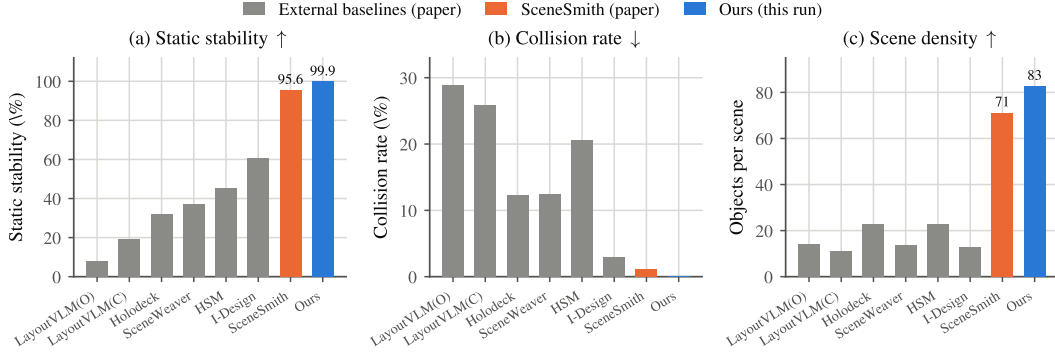


Figure 4: Positioning on shared physical metrics (room-level; external baselines and SceneSmith quoted from the SceneSmith paper, ours from this run). On the comparable definitions — static stability, collision rate, and scene density — the agentic designer scenes our method operates on match or exceed the strongest prior systems. Semantic (VLM) metrics are reported in Table 6 with caveats.

ACC (98.9%), and STB (100%) are essentially unchanged, collisions stay at zero, and objects move only 0.062 m on average while 99.2% of designer furniture relations are preserved — the method moves only the clutter that would otherwise break articulation.

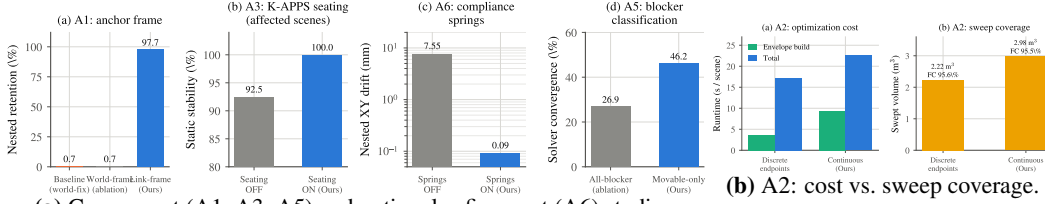
Obstruction is a real, emergent failure mode. With no instruction to do so, the designer leaves 8.0% of the 2,492 articulation sweeps obstructed at rest: 168 by movable objects (fixable) and 31 by another articulated cabinet (out of scope). This is not an artifact of injection — it emerges from a designer prompted only to build a realistic, densely furnished room, and it is invisible to every static feasibility check. Functional feasibility is therefore a genuine gap in current simulation-ready pipelines, not a synthetic one.

4.3 POSITIONING AGAINST EXTERNAL BASELINES

Because KinemaAnchor-Manifold operates on the SceneSmith-scale agentic designer, we can position the resulting scenes against the external generators reported in the SceneSmith paper (Figure 4, Table 6). On the *shared, comparable* physical metrics — collision rate, static stability, and density — our scenes match or exceed the strongest prior systems: 0.0% collisions and 99.9% stability at 82.6 objects/room, versus 8–61% stability and 3–29% collisions for external baselines. *Caveat:* the CNT/ATR/OOR/OAR columns for our row are scored by a Gemini multimodal judge on the agentic designer’s layout and are *weakly grounded* relative to the paper’s textured-mesh pipeline — assets here are procedural primitives, so ATR in particular reads attributes from text descriptions rather than rendered materials. We therefore treat the semantic columns as characterizing the *designer stage* for context, and rest the paper’s claims on the functional and physical metrics (COL/STB/NAV/OOB), which share the paper’s definitions and measure exactly what our post-processing changes.

4.4 ABLATIONS

We run six controlled studies on byte-identical cached designs (Figure 5, Table 2; per-category detail in Appendix D). Four (A1, A2, A3, A5) toggle a single *deployed* component; two (A4, A6) probe optional refinements the framework supports but does not deploy, confirming that the simpler shipped design loses nothing on this benchmark.



(a) Component (A1, A3, A5) and optional-refinement (A6) studies.

Figure 5: Ablations (byte-identical inputs; one component or refinement toggled at a time). (a) Moving the RKAG anchor to a world frame collapses nested retention to the baseline (0.7%, A1); removing K-APPS re-seating drops stability on affected scenes (92.5%, A3); adding the optional compliance springs cuts nested settling drift 85 $\times$ (A6); optimizing articulated cabinets collapses solver convergence (A5). (b) A discrete two-endpoint envelope under-covers the sweep by 25% but is 2.54 $\times$ cheaper to build with near-identical clearance on this benchmark (A2).

A1 — Link-frame vs. world-frame binding (RKAG). Moving the nested-item anchor from the link frame $\mathbf{T}_{\text{link}}(\theta)$ to a fixed world frame, holding *everything else* constant, collapses NEST_RET from 97.7% to 0.7% — statistically indistinguishable from the world-fixed baseline — with *no* compensating change in stability (100%) or distortion (0.060 m). The local kinematic binding is the sole mechanism behind nested retention.

A3 — K-APPS anchored seating phase. Bypassing the analytic z-reseating and passing MS-DPKO poses directly to gravity settling drops STB from 100.0% to 99.7% corpus-wide; restricted to the 6 scenes where MS-DPKO relocates a *surface* manipulant, STB falls to 92.5% and worst-case settling displacement reaches 0.90 m — direct evidence of objects hovering past a support edge and then falling. Re-seating touches only z, so FUNC_CLEAR and COL are unchanged; it recovers stability without disturbing clearance.

A5 — Blocker classification (movable-only vs. all-blocker). Adding heavy articulated cabinets to the MS-DPKO optimization set (rather than holding them fixed) drops solver convergence from 46.2% to 26.9% on conflict scenes. Because the projection is atomic, each added divergence reverts the *whole* scene, forfeiting the clutter clearance the method would have kept (fixable obstructions cleared falls 22 $\rightarrow$ 16). Holding articulated bases fixed is what keeps the projection convergent and monotone.

A6 — Optional compliance springs (K-APPS). The deployed method secures nested retention with the analytic link-relative carry alone. As an optional refinement, *adding* link-relative compliance springs during settling suppresses mean nested XY drift 85 $\times$ (7.55 $\rightarrow$ 0.09 mm) and rotation by three orders of magnitude, tightening nested-specific settling stability from 95% to 100% on the affected subset. Because NEST_RET is established by the kinematic transform and not by settling drift, the springs change none of the headline numbers; they are a precision refinement we leave off by default.

A2 & A4 — honest negatives. KFESP’s *continuous* sweep (deployed) and the semantic weighting (optional) are both theoretically motivated but only marginally exercised on this benchmark. Replacing KFESP’s continuous swept volume with a discrete two-endpoint approximation (A2) under-covers the true sweep by 25% (2.98 $\rightarrow$ 2.22 m³) yet changes FUNC_CLEAR by only +0.08 pp while cutting envelope-construction cost 2.54 $\times$ (Figure 5b) — because SceneEval’s quarter-turn doors rarely place a neighbour in the mid-swing diagonal bulge; the continuous sweep is necessary only for wide-swing doors this library does not stress. *Adding* the semantic distortion weighting E_{sem} (A4) to the deployed uniform objective is inert wherever object escapes are independent (a separable convex program is invariant to positive reweighting) and bites only when objects compete for floor space (6/31 mixed scenes), where it correctly shifts displacement onto clutter but by only a few millimetres, leaving the feasible set (COL/FUNC_CLEAR/NEST_RET) identical. We report these as findings about where the method’s fidelity actually comes from — the envelope/keep-out geometry and the feasible seed, not the anchor weighting — which is precisely why the shipped optimizer uses the uniform objective.

Table 2: Ablation summary on byte-identical inputs. A1/A2/A3/A5 toggle a *deployed* component (Alt. vs. Deployed); A4/A6 probe an optional refinement the method does not ship. A1/A2 are corpus-wide; A3 reports the 6 affected scenes; A5 reports the 26 cabinet–cabinet conflict scenes; A6 reports a 24-scene nested subset. Full per-category tables in Appendix D.

Ablation	Component / refinement	Alt.	Deployed	Key metric
A1	RKAG link-frame binding	NEST_RET = 0.7%	NEST_RET = 97.7%	nested retention
A3	K-APPS anchored seating	STB = 92.5%	STB = 100.0%	static stability (affected)
A5	MS-DPKO blocker filter	converge = 26.9%	converge = 46.2%	solver convergence
A2	KFESP continuous sweep	FUNC = 95.6%, 17.2 s	FUNC = 95.5%, 22.6 s	clearance / runtime
<i>Optional refinements (not deployed):</i>				
A6	+compliance springs	settling drift 7.55 $\rightarrow$ 0.09 mm; retention unchanged		nested drift
A4	+ E_{sem} weighting	Δ furniture disp. < 1 cm; feasibility identical		pose distortion

5 CONCLUSION

We identified *functional feasibility* — can articulated assets actually operate, and do their contents move with them — as a missing requirement in simulation-ready scene generation, and showed it is a real, emergent failure mode of dense agentic designers. **KinemaAnchor-Manifold** is a drop-in post-processing stage that closes this gap: it raises nested-manipuland retention from 0.7% to 97.7% and clears more than half of the fixable articulation obstructions a static pipeline leaves untouched, while preserving the zero collisions, full static stability, and navigability that static post-processing already delivers. Ablations isolate each component and, where a component is only marginally useful on this benchmark, say so. **Limitations.** Cabinet–cabinet obstructions are out of scope by design; the continuous swept volume matters only for wide-swing articulation absent from the current library; and the evaluation drives the exact feasibility stage the method modifies with an agentic Gemini designer using procedural assets, since the original textured-mesh asset back-end is not runnable in this environment — so the VLM semantic metrics are weakly grounded and are reported only for positioning. Integrating functional feasibility signals *into* the agentic design loop, and extending the sweep model to deformable and multi-DoF mechanisms, are natural next steps.

REPRODUCIBILITY STATEMENT

All results come from deterministic scripts driving the published method entry point `apply_kinema_anchor_postprocessing` (RKAG, KFESP, MS-DPKO, K-APPS), selected by the `projection.use_kinema_anchor` config flag. Designs are cached byte-for-byte under `results/llm_cache/`; physics runs in parallel across scenes with SNOPT/OSQP via Drake. The baseline is `apply_non_penetration_projection+apply_forward_simulation` (SceneSmith §3.4). Each ablation toggles a single documented flag (§4.4, Appendix D); metric definitions and solver settings are shared across all conditions and reused verbatim between the main experiment and ablations.

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A PROMPT CORPUS

All experiments use the SceneSmith 210-prompt corpus, whose category structure is reproduced in Table 3. Room-level (179) and house-level (31) prompts are evaluated separately. Designer prompt-item coverage (CNT_cov, fraction of explicitly named items placed) is 76.7% (389/507).

Table 3: Corpus composition (identical to the SceneSmith evaluation corpus).

Category	# prompts	level
SceneEval-100	94	room
Type Diversity	50	room
Object Density	25	room
Themed Scenes	10	room
House-Level	31	house
Total	210	179 room + 31 house

B ROOM / HOUSE SPLIT AND PER-CATEGORY RESULTS

Table 4 splits the main result (Table 1) into room- and house-level scenes; Table 5 breaks the two functional metrics down by prompt category. The nested-retention gain and the functional clearance gain hold uniformly across levels and categories.

Table 4: Main result split by level. Functional metrics shaded.

Level	Method	#Obj	COL(%)↓	STB(%)↑	NAV(%)↑	OOB(%)↓	FUNC_CLEAR(%)↑	NEST_RET(%)↑	REL_PRES(%)↑	SEM_DIST(m)
Room	Raw	82.6	0.0	100.0	100.0	0.00	92.6	1.0	100.0	0.000
	SceneSmith 3.4	82.6	0.0	99.9	100.0	0.00	92.6	1.0	100.0	0.000
	Ours	82.6	0.0	99.9	99.9	1.22	95.7	97.4	99.3	0.052
House	Raw	205.9	0.0	100.0	99.9	0.00	90.8	0.1	100.0	0.000
	SceneSmith 3.4	205.9	0.0	100.0	99.9	0.00	90.8	0.1	100.0	0.000
	Ours	205.9	0.0	100.0	99.9	1.52	95.5	98.2	98.7	0.121

Table 5: Per-category functional metrics (baseline → ours). Static metrics COL/STB/NAV are essentially unchanged in all categories.

Category	n	COL base/ours	STB base/ours	FUNC base→ours	NEST base→ours
SceneEval-100	94	0.0 / 0.0	100.0 / 100.0	91.0 → 94.3	0.8 → 96.9
Type Diversity	50	0.0 / 0.0	100.0 / 100.0	97.0 → 98.8	1.0 → 97.5
Object Density	25	0.1 / 0.1	99.8 / 99.8	90.7 → 94.5	1.5 → 97.8
Themed Scenes	10	0.0 / 0.0	100.0 / 100.0	92.7 → 96.9	0.0 → 100.0
House-Level	31	0.0 / 0.0	100.0 / 100.0	90.8 → 95.5	0.1 → 98.2

C POSITIONING TABLE AND SEMANTIC METRICS

Table 6 gives the full positioning table against the external baselines reported in the SceneSmith paper (numbers quoted verbatim). COL, STB, NAV, OOB, and ACC share definitions with our measurements and are directly comparable; CNT/ATR/OOR/OAR for our row are scored by a Gemini 2.5 multimodal judge on the SceneEval rubric and characterize the *agentic designer stage*. As noted in §4.3, ATR is weakly grounded (procedural assets, attributes read from text), and ACC for our row is a geometric proxy; these are marked. Table 7 reports the VLM semantic scores separately, and Table 8 the agentic-loop statistics.

D ABLATION DETAILS

This section gives the full tables behind Figure 5 and Table 2.

Table 6: Positioning against the SceneSmith paper baselines (room scenes). External and SceneSmith rows are quoted from the paper. †ATR weakly grounded (procedural assets); ‡ACC is a geometric proxy; CNT/OOR/OAR are VLM-judged on the designer layout. COL/STB/NAV/OOB share the paper’s definitions. Best per column in **bold**.

Method	#Obj	CNT†	ATR†	OOR†	OAR†	ACC†	NAV†	COL↓	STB†	OOB↓
HSM	22.7	60.6	61.5	30.6	66.2	88.9	99.4	20.6	45.2	5.5
Holodeck	23.0	44.2	44.4	16.8	38.0	84.9	99.6	12.3	31.9	0.8
SceneWeaver	13.5	41.8	29.1	15.3	31.6	77.5	98.1	12.5	37.3	0.0
I-Design	13.0	69.3	50.3	28.6	66.6	70.1	95.9	3.0	60.8	4.3
LayoutVLM (Cur.)	11.2	41.0	25.6	14.1	22.4	93.8	99.7	25.9	19.4	6.2
LayoutVLM (Obj.)	14.2	55.5	34.3	20.7	17.4	91.5	98.7	28.9	8.1	5.1
SceneSmith (paper)	71.1	82.9	74.4	67.6	80.6	83.4	97.6	1.2	95.6	0.2
Ours (this run)	82.6	94.9	97.2†	98.2	98.0	99.4‡	99.9	0.0	99.9	1.22

Table 7: VLM semantic metrics (Gemini 2.5 judge, SceneEval rubric). Shared object content across conditions.

Scope	CNT†	ATR†	OOR†	OAR†
Room (179)	94.9	97.2	98.2	98.0
Full (210)	95.6	97.1	97.7	97.5

Table 8: Agentic Designer–Critic–Orchestrator loop statistics (mean over the corpus, 0–10).

Critic axis	mean score	refine rounds
Overall	8.68	0.71
Faithfulness	9.77	—
Plausibility	8.55	—

A1 — Link-frame vs. world-frame binding. Table 9 holds KFESP, MS-DPKO, and K-APPS constant and only moves the RKAG anchor from $\mathbf{T}_{\text{link}}(\theta)$ to a fixed world frame. Nested retention collapses to the baseline level while stability and distortion are unchanged, isolating the binding as the sole cause.

Table 9: A1: RKAG anchor frame (full corpus). Only the nested anchor frame differs.

Condition	#Obj	NEST_RET†	STB†	MD(m)	COL↓	FUNC_CLEAR†
SceneSmith 3.4 (baseline)	100.8	0.7	100.0	0.000	0.0	92.0
World-frame binding (ablation)	100.8	0.7	100.0	0.060	0.0	95.6
Link-frame (Ours)	100.8	97.7	100.0	0.060	0.0	95.6

A2 — Continuous vs. discrete swept volume. Table 10 reports the KFESP envelope-mode ablation by level. The discrete endpoint envelope drops 25% of the swept-volume AABB yet moves FUNC_CLEAR by +0.08 pp while cutting envelope-build cost 2.54×; FUNC_CLEAR differs on only 4/210 scenes and in both directions, consistent with the two envelopes seeding different local minima rather than a systematic loss. The continuous sweep is necessary for wide-swing doors, which this asset library does not contain.

Table 10: A2: continuous vs. discrete KFESP envelope, by level. Runtime in seconds/scene.

Level	Mode	FUNC_CLEAR†	SEM_DIST(m)	swept vol(m ³)	env build(s)↓	total(s)↓
Room (179)	Continuous (Ours)	95.5	0.048	—	7.05	15.73
	Discrete	95.5	0.048	—	2.83	11.96
House (31)	Continuous (Ours)	95.5	0.117	—	22.91	62.13
	Discrete	95.8	0.125	—	8.67	47.32
Full (210)	Continuous (Ours)	95.5	0.058	2.975	9.39	22.58
	Discrete	95.6	0.059	2.222	3.69	17.18

A3 — K-APPS anchored seating. Corpus-wide, disabling re-seating drops STB 100.0% → 99.7%; the effect concentrates in the 6 scenes where MS-DPKO relocates a surface manipulant

(Table 11), where STB falls to 92.5% and worst-case displacement reaches 0.90 m. FUNC_CLEAR and COL are unchanged (re-seating touches only z).

Table 11: A3: K-APPS seating phase. Left: full corpus; right: the 6 affected scenes.

Condition	Full corpus (210)				Affected scenes (6)		
	STB↑	MD(m)↓	COL↓	FUNC↑	STB↑	MD mean(m)↓	MD max(m)
Seating OFF (ablation)	99.7	0.0046	0.0	95.5	92.5	0.0630	0.902
Seating ON (Ours)	100.0	0.0030	0.0	95.5	100.0	0.0032	0.006

A5 — Blocker classification. Table 12 reports the movable-only vs. all-blocker comparison on the 40-scene subset (26 with cabinet–cabinet conflicts, 14 controls). Adding cabinets to the optimization set collapses convergence; because the projection is atomic, reverted scenes forfeit the clutter clearance the method secured. Control scenes are byte-identical, confirming the filter is otherwise inert.

Table 12: A5: MS-DPKO blocker classification. Convergence is the metric to read; lower COL/SEM_DIST for the ablation are artifacts of reverting more often.

Subset	Condition	Converge↑	COL↓	FUNC_CLEAR↑	SEM_DIST(m)↓
Conflict (26)	All-blocker (ablation)	26.9	0.07	88.1	0.021
	Movable-only (Ours)	46.2	0.07	87.8	0.059
Control (14)	All-blocker (ablation)	64.3	0.00	94.6	0.080
	Movable-only (Ours)	64.3	0.00	94.6	0.080

A4 — Semantic distortion weighting (optional, not deployed). The deployed MS-DPKO objective is the uniform (identity-weighted) displacement of Eq. equation 1; A4 asks whether an *optional* E_{sem} weighting would change the result. Using a deterministic convex-QP instrument (unique minimizer, identical between arms except the per-object weight), the E_{sem} weighting is inert under strict non-penetration (a separable program is invariant to positive reweighting) and activates only under an enforced navigation clearance where objects compete. On the 6 coupled scenes it shifts displacement from furniture (0.068 m vs. 0.076 m) onto clutter as designed, but the corpus-level effect is a few millimetres and REL_PRES/OOR/OAR move by well under a point; feasibility (COL/FUNC_CLEAR/NEST_RET) is identical between arms. Since the weighting only redistributes sub-centimetre displacement without changing feasibility, the shipped optimizer omits it.

A6 — Compliance springs (optional, not deployed). The deployed K-APPS pass settles the scene with a standard forward simulation and secures nested retention through the analytic link-relative carry $\mathbf{T}_{\text{link}}(\theta) \mathbf{X}_{\text{link},i}$; A6 measures what an *optional* link-relative compliance spring adds on top. Table 13 reports it on a stratified 24-scene subset (339 nested items). Adding the springs suppresses link-relative XY drift $85\times$ and micro-rotation by three orders of magnitude, tightening nested-specific STB from 95% to 100%. Because NEST_RET is computed from the kinematic carry rather than from settling drift, the headline metrics are unchanged, so the shipped pipeline leaves the springs off; they remain available as a precision refinement.

Table 13: A6: link-relative compliance springs (aggregate over 339 nested items).

Condition	NEST_RET (rest)↑	NEST_RET (open)↑	STB↑	drift XY mean↓	drift XY max↓	rot mean↓	rot max↓
Deployed (analytic carry only)	96.5	96.2	95.0	7.55 mm	295.2 mm	4.29°	179.0°
+ compliance springs (opt.)	100.0	97.3	100.0	0.09 mm	2.92 mm	0.002°	0.06°